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Equation Reference for stellar2d

All numbered equations in this document are referenced by source code comments of the form `// Eq. (X.Y)`. This is the authoritative specification — if code disagrees with this document, the code has a bug.

Coordinate system: axisymmetric spherical (r, θ) , with $\partial/\partial\phi = 0$ and $v_\phi = 0$. $\theta \in [0, \pi]$ is the polar angle measured from the north pole.

§1 Governing Equations

Conservative variables

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho v_r \\ \rho v_\theta \\ \rho E \end{pmatrix} \quad (1.1)$$

where total specific energy $E = e + \frac{1}{2}(v_r^2 + v_\theta^2)$ and e is specific internal energy.

Ideal gas equation of state

$$P = (\gamma - 1) \rho e \quad (1.2)$$

Sound speed

$$c_s = \sqrt{\gamma P / \rho} \quad (1.3)$$

Euler equations in conservative form (spherical, axisymmetric)

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{1}{r^2} \frac{\partial(r^2 \mathbf{F})}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta \mathbf{G})}{\partial \theta} = \mathbf{S} \quad (1.4)$$

Radial flux:

$$\mathbf{F} = \begin{pmatrix} \rho v_r \\ \rho v_r^2 + P \\ \rho v_r v_\theta \\ (\rho E + P) v_r \end{pmatrix} \quad (1.5)$$

Theta flux:

$$\mathbf{G} = \begin{pmatrix} \rho v_\theta \\ \rho v_r v_\theta \\ \rho v_\theta^2 + P \\ (\rho E + P) v_\theta \end{pmatrix} \quad (1.6)$$

Geometric + gravity source:

$$\mathbf{S} = \begin{pmatrix} 0 \\ \frac{\rho v_\theta^2}{r} + \frac{2P}{r} - \rho \frac{\partial \Phi}{\partial r} \\ \frac{P \cot \theta}{r} - \frac{\rho v_r v_\theta}{r} - \frac{\rho}{r} \frac{\partial \Phi}{\partial \theta} \\ -\rho \mathbf{v} \cdot \nabla \Phi \end{pmatrix} \quad (1.7)$$

Note: the $2P/r$ and $P \cot \theta / r$ terms are geometric consequences of expressing the divergence in spherical coordinates, not additional physics.

Poisson equation (self-gravity)

$$\nabla^2 \Phi = 4\pi G \rho \quad (1.8)$$

Expanded in axisymmetric spherical coordinates:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) = 4\pi G \rho \quad (1.9)$$

§2 Finite Volume Discretization

Structured grid with $N_r \times N_\theta$ cells. Indices: $i = 0, \dots, N_r - 1$ (radial), $j = 0, \dots, N_\theta - 1$ (polar). Cell interfaces at $r_{i+1/2}$, $\theta_{j+1/2}$.

Logarithmic radial mesh

$$r_{i+1/2} = R_{\text{outer}} \left(\frac{i}{N_r} \right)^\alpha \quad (2.1)$$

Cell volume (azimuthal 2π factor omitted; cancels in all FV updates)

$$V_{ij} = \frac{r_{i+1/2}^3 - r_{i-1/2}^3}{3} (\cos \theta_{j-1/2} - \cos \theta_{j+1/2}) \quad (2.2)$$

Radial face area

$$A_{i+1/2,j}^r = r_{i+1/2}^2 (\cos \theta_{j-1/2} - \cos \theta_{j+1/2}) \quad (2.3)$$

Theta face area

$$A_{i,j+1/2}^\theta = \sin \theta_{j+1/2} \frac{r_{i+1/2}^2 - r_{i-1/2}^2}{2} \quad (2.4)$$

Semi-discrete finite volume update

$$\frac{d\mathbf{U}_{ij}}{dt} = -\frac{1}{V_{ij}} \left[A_{i+1/2,j}^r \hat{\mathbf{F}}_{i+1/2,j} - A_{i-1/2,j}^r \hat{\mathbf{F}}_{i-1/2,j} + A_{i,j+1/2}^\theta \hat{\mathbf{G}}_{i,j+1/2} - A_{i,j-1/2}^\theta \hat{\mathbf{G}}_{i,j-1/2} \right] + \mathbf{S}_{ij} \quad (2.5)$$

§3 MUSCL Reconstruction

Minmod limiter

$$\text{minmod}(a, b) = \begin{cases} \text{sgn}(a) \min(|a|, |b|) & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

Van Leer limiter

$$\text{vanleer}(a, b) = \begin{cases} \frac{2ab}{a+b} & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.2)$$

Limited slope at cell i

$$\sigma_i = \text{limiter}(w_i - w_{i-1}, w_{i+1} - w_i) \quad (3.3)$$

Reconstructed left state at face $i + 1/2$

$$w_{i+1/2}^L = w_i + \frac{1}{2} \sigma_i \quad (3.4)$$

Reconstructed right state at face $i + 1/2$

$$w_{i+1/2}^R = w_{i+1} - \frac{1}{2} \sigma_{i+1} \quad (3.5)$$

§4 HLLC Riemann Solver

Reference: Toro, “Riemann Solvers and Numerical Methods for Fluid Dynamics”, 3rd ed., §10.4.

Wave speed estimates

$$S_L = \min(u_L - c_L, u_R - c_R), \quad S_R = \max(u_L + c_L, u_R + c_R) \quad (4.1)$$

Contact wave speed

$$S^* = \frac{P_R - P_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)} \quad (4.2)$$

Total energy density

$$\mathcal{E} = \frac{P}{\gamma - 1} + \frac{1}{2} \rho (u^2 + v_t^2) \quad (4.3)$$

Physical flux (in the normal direction u , tangential v_t)

$$\mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + P \\ \rho u v_t \\ (\mathcal{E} + P) u \end{pmatrix} \quad (4.4)$$

For radial faces: $u = v_r$, $v_t = v_\theta$. For theta faces: $u = v_\theta$, $v_t = v_r$.

HLLC star-state density

$$\rho_K^* = \rho_K \frac{S_K - u_K}{S_K - S^*} \quad (4.5)$$

HLLC star-state energy

$$\mathcal{E}_K^* = \rho_K^* \left[\frac{\mathcal{E}_K}{\rho_K} + (S^* - u_K) \left(S^* + \frac{P_K}{\rho_K (S_K - u_K)} \right) \right] \quad (4.6)$$

HLLC numerical flux

$$\hat{\mathbf{F}} = \begin{cases} \mathbf{F}_L & \text{if } S_L \geq 0 \\ \mathbf{F}_L + S_L (\mathbf{U}_L^* - \mathbf{U}_L) & \text{if } S_L < 0 \leq S^* \\ \mathbf{F}_R + S_R (\mathbf{U}_R^* - \mathbf{U}_R) & \text{if } S^* < 0 \leq S_R \\ \mathbf{F}_R & \text{if } S_R < 0 \end{cases} \quad (4.7)$$

§5 Geometric Source Terms

These arise from expanding $\nabla \cdot$ in spherical coordinates. They must be discretized in a **volume-consistent** manner to achieve well-balanced equilibrium (see §5.3).

Radial momentum geometric source (continuous)

$$S_{m_r}^{\text{geom}} = \frac{\rho v_\theta^2}{r} + \frac{2P}{r} \quad (5.1)$$

Theta momentum geometric source (continuous)

$$S_{m_\theta}^{\text{geom}} = \frac{P \cot \theta}{r} - \frac{\rho v_r v_\theta}{r} \quad (5.2)$$

Volume-consistent discrete geometric source (well-balanced)

The continuous $2P/r$ source must be discretized consistently with the FV divergence operator. We replace the point-value $2P/r$ with its volume average:

$$\left\langle \frac{2P}{r} \right\rangle_{ij} = \frac{1}{V_{ij}} \int_{\text{cell}} \frac{2P}{r} r^2 \sin \theta dr d\theta = \frac{P_{ij}}{V_{ij}} (r_{i+1/2}^2 - r_{i-1/2}^2) (\cos \theta_{j-1/2} - \cos \theta_{j+1/2}) \quad (5.3)$$

(assuming P approximately uniform within the cell).

Similarly, the $P \cot \theta / r$ source:

$$\left\langle \frac{P \cot \theta}{r} \right\rangle_{ij} = \frac{P_{ij}}{V_{ij}} \frac{r_{i+1/2}^2 - r_{i-1/2}^2}{2} (\sin \theta_{j-1/2} - \sin \theta_{j+1/2}) \quad (5.4)$$

(derived from $\int \cos \theta d\theta = \sin \theta$).

§6 Gravity

Gravity source — radial momentum

$$S_{m_r}^g = -\rho \frac{\partial \Phi}{\partial r} \quad (6.1)$$

Gravity source — theta momentum

$$S_{m_\theta}^g = -\frac{\rho}{r} \frac{\partial \Phi}{\partial \theta} \quad (6.2)$$

Gravity source — energy

$$S_E^g = -\rho \left(v_r \frac{\partial \Phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial \Phi}{\partial \theta} \right) \quad (6.3)$$

Discrete Laplacian — radial part

$$L_{ij}^r = \frac{1}{r_i^2} \frac{r_{i+1/2}^2 \frac{\Phi_{i+1,j} - \Phi_{ij}}{\delta r_i^+} - r_{i-1/2}^2 \frac{\Phi_{ij} - \Phi_{i-1,j}}{\delta r_i^-}}{\Delta r_i} \quad (6.4)$$

where $\delta r_i^+ = r_{i+1}^c - r_i^c$, $\delta r_i^- = r_i^c - r_{i-1}^c$, $\Delta r_i = r_{i+1}^f - r_i^f$ (face-to-face width).

Discrete Laplacian — theta part

$$L_{ij}^\theta = \frac{1}{r_i^2 \sin \theta_j} \frac{\sin \theta_{j+1/2} \frac{\Phi_{i,j+1} - \Phi_{ij}}{\delta \theta_j^+} - \sin \theta_{j-1/2} \frac{\Phi_{ij} - \Phi_{i,j-1}}{\delta \theta_j^-}}{\Delta \theta_j} \quad (6.5)$$

where $\delta \theta_j^\pm$ are center-to-center distances, $\Delta \theta_j$ is face-to-face width.

Poisson boundary conditions

Dirichlet at outer boundary (monopole approximation):

$$\Phi(R_{\text{outer}}) = -\frac{G M_{\text{total}}}{R_{\text{outer}}} \quad (6.6)$$

Neumann at center ($r = 0$, symmetry):

$$\left. \frac{\partial \Phi}{\partial r} \right|_{r=0} = 0 \quad (6.7)$$

Neumann at poles ($\theta = 0, \pi$, axial symmetry):

$$\left. \frac{\partial \Phi}{\partial \theta} \right|_{\theta=0,\pi} = 0 \quad (6.8)$$

Gravity gradient at cell center (face-based interpolation)

Face gradients:

$$\left. \frac{\partial \Phi}{\partial r} \right|_{i+1/2} = \frac{\Phi_{i+1,j} - \Phi_{ij}}{r_{i+1}^c - r_i^c} \quad (6.9a)$$

Cell-center gradient by weighted average of flanking face gradients:

$$\left. \frac{\partial \Phi}{\partial r} \right|_i = \frac{\delta r_i^+ \cdot g_{i-1/2} + \delta r_i^- \cdot g_{i+1/2}}{\delta r_i^+ + \delta r_i^-} \quad (6.9b)$$

At $i = 0$: use $g_{i-1/2} = 0$ (Neumann, Eq. 6.7). At $i = N_r - 1$: use the Dirichlet value (Eq. 6.6) to form $g_{i+1/2}$.

Theta gradient analogously:

$$\left. \frac{\partial \Phi}{\partial \theta} \right|_{j+1/2} = \frac{\Phi_{i,j+1} - \Phi_{ij}}{\theta_{j+1}^c - \theta_j^c} \quad (6.10a)$$

$$\left. \frac{\partial \Phi}{\partial \theta} \right|_j = \frac{\delta \theta_j^+ \cdot g_{j-1/2}^\theta + \delta \theta_j^- \cdot g_{j+1/2}^\theta}{\delta \theta_j^+ + \delta \theta_j^-} \quad (6.10b)$$

At $j = 0$ and $j = N_\theta - 1$: use $g^\theta = 0$ (Neumann, Eq. 6.8).

§7 Time Integration

RK2 — Heun's method

Stage 1:

$$\mathbf{U}^* = \mathbf{U}^n + \Delta t \mathbf{R}(\mathbf{U}^n) \quad (7.1)$$

Stage 2:

$$\mathbf{U}^{**} = \mathbf{U}^* + \Delta t \mathbf{R}(\mathbf{U}^*) \quad (7.2)$$

Average:

$$\mathbf{U}^{n+1} = \frac{1}{2}(\mathbf{U}^n + \mathbf{U}^{**}) \quad (7.3)$$

where $\mathbf{R}(\mathbf{U})$ includes flux divergence, geometric source, and gravity source.

CFL condition

$$\Delta t = C_{\text{CFL}} \cdot \min_{i,j} \left[\min \left(\frac{\Delta r_i}{|v_r| + c_s}, \frac{r_i \Delta \theta_j}{|v_\theta| + c_s} \right) \right] \quad (7.4)$$

§8 Boundary Conditions

Inner boundary ($r = 0$): reflecting symmetry

For ghost cell at layer g ($g = 1, 2, \dots$):

$$\rho_{-g,j} = \rho_{g-1,j}, \quad (\rho v_r)_{-g,j} = -(\rho v_r)_{g-1,j}, \quad (\rho v_\theta)_{-g,j} = (\rho v_\theta)_{g-1,j}, \quad (\rho E)_{-g,j} = (\rho E)_{g-1,j} \quad (8.1)$$

Outer boundary ($r = R$): zero-gradient outflow

$$\mathbf{U}_{N_r+g,j} = \mathbf{U}_{N_r-1,j} \quad (8.2)$$

Axis boundaries ($\theta = 0$ and $\theta = \pi$): reflecting symmetry in v_θ

For $\theta = 0$ (north pole), ghost layer g :

$$\rho_{i,-g} = \rho_{i,g-1}, \quad (\rho v_r)_{i,-g} = (\rho v_r)_{i,g-1}, \quad (\rho v_\theta)_{i,-g} = -(\rho v_\theta)_{i,g-1}, \quad (\rho E)_{i,-g} = (\rho E)_{i,g-1} \quad (8.3)$$

South pole ($\theta = \pi$) identical with indices mirrored from $j = N_\theta - 1$.

§9 Initial Conditions

Lane-Emden equation

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\Theta}{d\xi} \right) + \Theta^n = 0, \quad \Theta(0) = 1, \quad \Theta'(0) = 0 \quad (9.1)$$

Lane-Emden length scale

$$\alpha^2 = \frac{(n+1) K \rho_c^{1/n-1}}{4\pi G} \quad (9.2)$$

Stellar radius

$$R_\star = \alpha \xi_1 \quad (9.3)$$

where ξ_1 is the first zero of $\Theta(\xi)$.

Density from Lane-Emden solution

$$\rho(r) = \rho_c \Theta\left(\frac{r}{\alpha}\right)^n \quad (9.4)$$

Polytropic pressure

$$P = K \rho^{1+1/n} \quad (9.5)$$

Sedov blast energy deposition

$$P_{\text{blast}} = (\gamma - 1) \rho_0 \frac{E_{\text{blast}}}{V_{\text{blast}}} \quad (9.6)$$

where V_{blast} uses the same volume convention as the code (no 2π azimuthal factor).

Evrard collapse density profile

$$\rho(r) = \frac{M}{2\pi R^3} \frac{1}{r}, \quad r < R \quad (9.7)$$

Jeans instability perturbation

$$\rho = \rho_0 [1 + \epsilon \cos(k_r r) \cos(k_\theta \theta)] \quad (9.8)$$

§10 Low-Mach Implicit Solver (JFNK)

The low-Mach solver uses Backward Euler with Jacobian-Free Newton-Krylov (JFNK) to take time steps unconstrained by the acoustic CFL.

Backward Euler nonlinear system

$$F(U^{n+1}) = \frac{U^{n+1} - U^n}{\Delta t} - R(U^{n+1}) = 0 \quad (10.1)$$

where $R(U)$ is the spatial residual (§1, §5, §6) using 1st-order upwind advection and central-difference pressure/gravity gradients.

Newton iteration

$$J_k \delta U = -F(U^k), \quad U^{k+1} = U^k + \alpha \delta U \quad (10.2)$$

with backtracking line search on the scaled merit function $\|L^{-1}F\|_2$.

Jacobian-free matrix-vector product

$$J v \approx \frac{F(U^k + \varepsilon v) - F(U^k)}{\varepsilon}, \quad \varepsilon = \sqrt{\epsilon_{\text{mach}}} \frac{1 + \|U\|}{\|v\|} \quad (10.3)$$

FGMRES linear solver

Right-preconditioned FGMRES(120) with Eisenstat-Walker adaptive forcing (initial $\eta = 10^{-2}$, EW Choice 2).

1D radial gravity

Gravity is computed from angle-averaged density via cumulative mass integral (no Poisson solve, no GMG noise):

$$g_r(r_i) = -\frac{G M(< r_i)}{r_i^2}, \quad M(< r_i) = \sum_{k < i} \bar{\rho}_k V_k \quad (10.4)$$

where $\bar{\rho}_k$ is the θ -averaged density at shell k .

Well-balanced residual (reference-state subtraction)

The momentum residual uses perturbation form to achieve $R(U_{\text{HSE}}) = 0$ to machine precision:

$$F_r = -\nabla P' + \rho' g_0(r) + \rho g'(r) + S_{\text{geom}} \quad (10.5)$$

where primed quantities are deviations from the HSE reference state.

Atmosphere treatment

Cells with $\rho_0 < 10^{-6} \rho_{\text{max}}$ are treated as atmosphere: $R = 0$ in the residual kernel, so Newton does not evolve them.

§11 Physics-Based Preconditioner (PBP)

The JFNK preconditioner $M \approx J$ uses the block structure of the Jacobian to incorporate a pressure Poisson solve, enabling GMRES convergence in $O(10)$ iterations regardless of Δt .

Jacobian block structure

The 4-DOF Jacobian has the block form:

$$J = \begin{pmatrix} A_\rho & 0 & 0 & 0 \\ g_0 & A_{vr} & 0 & B_r \\ 0 & 0 & A_{v\theta} & B_\theta \\ 0 & C_r & C_\theta & A_E \end{pmatrix} \quad (11.1)$$

where $A_v = -(1/\Delta t + \text{advection rate})$ (diagonal-dominant), $B = -(\gamma-1)\nabla$ (pressure gradient), and $C = -P \nabla \cdot (1/\rho)$ (compression work).

Schur complement

Eliminating velocity from the energy equation gives the pressure Schur complement:

$$S = A_E - C A_v^{-1} B \approx \nabla \cdot \left(\frac{\gamma-1}{A_p} \nabla \right) \quad (11.2)$$

which is a variable-coefficient Poisson operator, solved by geometric multigrid (GMG).

PBP preconditioner application ($y = M^{-1}x$)

Given input $x = (x_\rho, x_{mr}, x_{m\theta}, x_E)$:

Step 1 – Momentum predict (2-DOF block-tridiagonal r-line solve):

$$\tilde{v}_r, \tilde{v}_\theta = A_v^{-1} (x_{mr}, x_{m\theta}) / \rho \quad (11.3)$$

Step 2 – Pressure Poisson (GMG, 3 V-cycles):

$$\nabla \cdot \left(\frac{1}{A_p} \nabla \delta p \right) = \nabla \cdot \tilde{v} \quad (11.4)$$

Step 3 – Velocity correct and assemble:

$$y_{mr} = \rho \left(\tilde{v}_r - \frac{1}{A_p} \frac{\partial \delta p}{\partial r} \right), \quad y_{m\theta} = \rho \left(\tilde{v}_\theta - \frac{1}{A_p} \frac{1}{r} \frac{\partial \delta p}{\partial \theta} \right) \quad (11.5)$$

$$y_\rho = (J^{-1})_{00} x_\rho, \quad y_E = (J^{-1})_{33} x_E \quad (11.6)$$

where $(J^{-1})_{00}$ and $(J^{-1})_{33}$ are diagonal elements of the block-Jacobi inverse (point Jacobi on ρ and energy).

Why this works

The preconditioner runs one cycle of the projection method *inside* the Krylov solver. The GMG pressure Poisson directly resolves the global pressure coupling that GMRES alone would need $O(N)$ iterations to discover. Because this is only a preconditioner, approximation errors do not affect the final solution — the outer JFNK iteration converges to the exact Backward Euler solution regardless of preconditioner quality.

§12 FAS Nonlinear Multigrid (Polar Grid)

The FAS (Full Approximation Scheme) solver operates on the same polar grid as the explicit solver (§1–§8), using nonlinear multigrid with block-Jacobi smoothing.

Well-balanced flux-level P_0 subtraction

The momentum flux divergence subtracts background pressure at each face to achieve $R(U_{\text{HSE}}) = 0$ to machine precision. For regular cells ($i \geq 1$):

$$(\nabla \cdot F)_{m_r} = -\frac{1}{V_{ij}} \left[A_{i+1/2}^r (\hat{F}_{i+1/2}^{m_r} - P_{i+1/2}^0) - A_{i-1/2}^r (\hat{F}_{i-1/2}^{m_r} - P_{i-1/2}^0) + A_{j+1/2}^\theta \hat{G}_{j+1/2}^{m_r} - A_{j-1/2}^\theta \hat{G}_{j-1/2}^{m_r} \right] \quad (12.1)$$

where $P_{i+1/2}^0 = \frac{1}{2}(P_0(r_i) + P_0(r_{i+1}))$ is the face-averaged HSE pressure. The theta momentum equation subtracts P^0 from the θ -face flux analogously.

Well-balanced gravity source (perturbation split)

$$S_{m_r}^g = \rho' g_{0,r} + \rho g'_r, \quad \rho' = \rho - \rho_0, \quad g' = g - g_0 \quad (12.2)$$

where ρ_0, g_0 are the HSE reference state. At equilibrium, $\rho' = 0$ and $g' = 0$, so the source vanishes exactly.

HSE defect subtraction

On coarse grids the discrete flux-level WB is not exact due to averaging. The solver precomputes and subtracts the frozen defect:

$$R_{\text{corrected}}(U) = R(U) - R(U_0) \quad (12.3)$$

ensuring $R_{\text{corrected}}(U_0) = 0$ on every grid level.

Origin cell ($i = 0$)

The innermost cell is a pie-slice wedge touching $r = 0$. It has no inner radial face (zero area). Geometry:

$$V_{0j} = \frac{1}{3} r_{1/2}^3 (\cos \theta_{j-1/2} - \cos \theta_{j+1/2}), \quad A_{1/2,j}^r = r_{1/2}^2 (\cos \theta_{j-1/2} - \cos \theta_{j+1/2}) \quad (12.4)$$

Volume-averaged $\langle 1/r \rangle$ for geometric source:

$$\left\langle \frac{1}{r} \right\rangle_{0j} = \frac{3}{2 r_{1/2}} \quad (12.5)$$

First-order HLLC is used at the origin (stencil too short for MUSCL). WB P_0 subtraction applies to the single outer radial face:

$$(\nabla \cdot F)_{m_r}|_{i=0} = -\frac{1}{V_{0j}} \left[A_{1/2}^r (\hat{F}_{1/2}^{m_r} - P_{1/2}^0) + A_{j+1/2}^\theta \hat{G}_{j+1/2}^{m_r} - A_{j-1/2}^\theta \hat{G}_{j-1/2}^{m_r} \right] \quad (12.6)$$

MUSCL reconstruction (perturbation form)

Slopes are computed on perturbation variables, then face states are restored using the face-centered HSE background:

$$\sigma_i^\rho = \text{limiter}(\rho'_i - \rho'_{i-1}, \rho'_{i+1} - \rho'_i) \quad (12.7a)$$

$$\rho_{i+1/2}^L = \rho_0(r_{i+1/2}) + \rho'_i + \frac{1}{2}\sigma_i^\rho, \quad \rho_{i+1/2}^R = \rho_0(r_{i+1/2}) + \rho'_{i+1} - \frac{1}{2}\sigma_{i+1}^\rho \quad (12.7b)$$

where $\rho_0(r_{i+1/2}) = \frac{1}{2}(\rho_{0,i} + \rho_{0,i+1})$ is the face-centered background. Pressure is treated identically.

FAS restrict (volume-weighted)

Fine-to-coarse state transfer with 2×2 coarsening:

$$U_{I,J}^H = \frac{\sum_{(i,j) \in \mathcal{C}(I,J)} U_{ij}^h V_{ij}^h}{\sum_{(i,j) \in \mathcal{C}(I,J)} V_{ij}^h} \quad (12.8)$$

where $\mathcal{C}(I, J)$ denotes the 2×2 fine children of coarse cell (I, J) .

FAS tau correction

The coarse-level FAS right-hand side incorporates the restricted fine-level defect so that the coarse solve drives the fine-level residual to zero:

$$\tau_H = \frac{U_H}{\Delta t} - R(U_H) + \bar{d}_H \quad (12.9)$$

where $\bar{d}_H$ is the restricted fine defect (Eq. 12.8 applied to $F(U^h) = R(U^h) - U^h/\Delta t + \tau^h$).

FAS prolongate (limited injection)

Coarse correction is piecewise-constant, clamped for safety:

$$\delta U_H = U_H^{\text{solved}} - U_H^{\text{restricted}} \quad (12.10a)$$

$$U_{ij}^h += \text{clamp}(\delta U_H, -\beta s_{ij}, +\beta s_{ij}), \quad \beta = 0.5 \quad (12.10b)$$

where s_{ij} is a local scale (ρ , ρc_s , or ρE depending on the equation).

Block-Jacobi smoother

$$U^{k+1} = U^k - \omega J_{\text{diag}}^{-1} F(U^k), \quad \omega = 0.8 \quad (12.11)$$

The 4×4 diagonal Jacobian block at each cell is:

$$J_{\text{diag}} = -\left(\frac{1}{\Delta t} + s_r\right)I + \text{off-diagonal couplings} \quad (12.12)$$

where $s_r = 2[(|v_r| + c_s)/\Delta r + (|v_\theta| + c_s)/(r\Delta\theta)]$ is the spectral radius estimate (factor 2 for MUSCL doubling). Off-diagonal entries capture pressure-energy coupling ($\partial(\nabla P)/\partial(\rho E)$) and gravity-density coupling ($\partial S^g/\partial\rho$). The block is inverted per cell via 4×4 Gauss-Jordan elimination.

Smooth floor

Density and internal energy are kept non-negative by a C^∞ smooth approximation to $\max(x, \varepsilon)$:

$$\text{floor}(x) = \frac{1}{2}(x + \sqrt{x^2 + 4\varepsilon^2}), \quad \varepsilon = 10^{-20} \quad (12.13)$$

Applied first to ρ , then to the internal energy $e_{\text{int}} = \rho E - \frac{1}{2}\rho v^2$.

CFL condition (polar grid)

$$\Delta t = C_{\text{CFL}} \min_{i,j} \left[\min \left(\frac{\Delta r_i}{|v_r| + c_s}, \frac{r_i \Delta\theta_j}{|v_\theta| + c_s} \right) \right] \quad (12.14)$$

Atmosphere cells ($\rho_0 < 10^{-6}\rho_{\text{max}}$) are excluded from the CFL computation.

§13 Strang Cartesian Solver

The Strang solver operates on a uniform Cartesian grid (x, y) , $[0, L_x] \times [0, L_y]$, with cell sizes $\Delta x = L_x/N_x$, $\Delta y = L_y/N_y$ and $n_g = 2$ ghost layers.

Perturbation storage

The solver stores perturbations from a 1-D isentropic HSE background:

$$\rho'_{ij} = \rho_{ij} - \bar{\rho}(y_j), \quad E'_{ij} = E_{ij} - \frac{\bar{p}(y_j)}{\gamma - 1} \quad (13.1)$$

Momentum $(\rho u, \rho v)$ is stored in total form (background velocity is zero).

Isentropic HSE background

$$\bar{\rho}(y) = \left[\rho_0^{\gamma-1} - \frac{(\gamma-1) g y}{\gamma K} \right]^{1/(\gamma-1)}, \quad \bar{p}(y) = K \bar{\rho}(y)^\gamma \quad (13.2)$$

where ρ_0 is the base density, K the polytropic constant, g the uniform gravitational acceleration, and γ the adiabatic index.

MC (Monotonised Central) limiter

$$\text{MC}(a, b) = \begin{cases} \text{sgn}(a) \min(\frac{1}{2}|a+b|, 2|a|, 2|b|) & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \quad (13.3)$$

Well-balanced y-sweep MUSCL reconstruction

Slopes are computed on perturbation quantities; face states are restored using face-centred HSE values:

$$\sigma_j^\rho = \text{MC}(\rho'_j - \rho'_{j-1}, \rho'_{j+1} - \rho'_j) \quad (13.4a)$$

$$\sigma_j^P = \text{MC}(P'_j - P'_{j-1}, P'_{j+1} - P'_j), \quad P'_j = P_j - \bar{p}(y_j) \quad (13.4b)$$

$$\rho_{j+1/2}^L = \bar{\rho}(y_{j+1/2}) + \rho'_j + \frac{1}{2}\sigma_j^\rho \quad (13.4c)$$

$$P_{j+1/2}^L = \bar{p}(y_{j+1/2}) + P'_j + \frac{1}{2}\sigma_j^P \quad (13.4d)$$

The x-sweep uses no WB (background is y-dependent only); slopes and reconstruction are on total variables.

MUSCL-Hancock predictor

Both left and right face states of a cell are evolved by half a time step using the within-cell flux difference:

$$\mathbf{U}_{\text{face}}^{n+1/2} = \mathbf{U}_{\text{face}}^n + \frac{\Delta t}{2\Delta x} [\mathbf{F}(\mathbf{W}_L) - \mathbf{F}(\mathbf{W}_R)] \quad (13.5)$$

where $\mathbf{W}_L, \mathbf{W}_R$ are the MUSCL-reconstructed left and right primitives of the same cell. In the y-sweep, the gravity source $-\rho g$ is included in the Hancock half-step to balance $\partial \bar{p} / \partial y$:

$$(\rho v)_{\text{face}}^{n+1/2} += -\frac{1}{2}\Delta t \rho g \quad (13.6)$$

Strang splitting

$$\mathbf{U}^{n+1} = \mathcal{L}_x\left(\frac{\Delta t}{2}\right) \circ \mathcal{L}_y\left(\frac{\Delta t}{2}\right) \circ \mathcal{L}_y\left(\frac{\Delta t}{2}\right) \circ \mathcal{L}_x\left(\frac{\Delta t}{2}\right) \mathbf{U}^n \quad (13.7)$$

Each $\mathcal{L}$ is a MUSCL-Hancock + LM-HLLC unsplit sweep in the corresponding direction.

Boundary conditions (Cartesian)

x-periodic (n_g ghost layers):

$$U_{-g,j} = U_{N_x-g,j}, \quad U_{N_x+g,j} = U_{g,j} \quad (13.8a)$$

y-bottom reflective ($y = 0$):

$$\rho_{i,-g} = \rho_{i,g-1}, \quad (\rho u)_{i,-g} = (\rho u)_{i,g-1}, \quad (\rho v)_{i,-g} = -(\rho v)_{i,g-1}, \quad E_{i,-g} = E_{i,g-1} \quad (13.8b)$$

y-top outflow (zero-gradient):

$$U_{i,N_y+g} = U_{i,N_y-1} \quad (13.8c)$$

CFL condition (Cartesian)

$$\Delta t = C_{\text{CFL}} \min_{i,j} \left[\min \left(\frac{\Delta x}{|u| + c_s}, \frac{\Delta y}{|v| + c_s} \right) \right] \quad (13.9)$$

§14 Low-Mach HLLC (LM-HLLC)

The LM-HLLC Riemann solver modifies the standard HLLC (§4) to reduce excessive numerical diffusion at low Mach number. The modification scales the pressure jump in the contact wave speed estimate.

Local Mach number

$$M_{\text{local}} = \frac{|u_L| + |u_R|}{c_L + c_R} \quad (14.1)$$

Blending factor

$$f(M) = \text{clamp}(M_{\text{local}}, M_{\text{cutoff}}, 1) \quad (14.2)$$

with $M_{\text{cutoff}} = 10^{-3}$. At $M \gg M_{\text{cutoff}}$, $f = 1$ and standard HLLC is recovered. At $M < M_{\text{cutoff}}$, $f = M_{\text{cutoff}}$ and the acoustic dissipation is suppressed.

Modified contact wave speed

In the Strang solver:

$$S^* = \frac{f(M)(P_R - P_L) + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)} \quad (14.3)$$

In the FAS solver (equivalent, alternative formulation):

$$P_{L,\text{mod}} = \bar{P} + \frac{1}{2}f(P_L - P_R), \quad P_{R,\text{mod}} = \bar{P} - \frac{1}{2}f(P_L - P_R) \quad (14.4a)$$

$$S^* = \frac{P_{R,\text{mod}} - P_{L,\text{mod}} + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)} \quad (14.4b)$$

where $\bar{P} = \frac{1}{2}(P_L + P_R)$. The two formulations are algebraically identical.

Physical interpretation

At low Mach number, the pressure jump across a contact wave is $O(M^2 P)$ while the acoustic pressure fluctuation is $O(M P)$. Standard HLLC treats both at the same order, creating $O(1/M)$ numerical dissipation. Scaling by $f(M)$ reduces the effective pressure jump, suppressing the spurious acoustic mode and allowing buoyancy-driven flows to develop correctly.

Consequence: sound waves with $M < M_{\text{cutoff}}$ are damped. For stellar convection ($M \sim 10^{-3}$ - 10^{-1}), this is acceptable because the dynamics are buoyancy-dominated.

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